Munekazu Ohno, Rainer Schmid–Fetzer
Clausthal University of Technology, Institute of Metallurgy, Clausthal-Zellerfeld, Germany

Thermodynamic assessment of Mg–Al–Mn phase equilibria, focusing on Mg-rich alloys

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A thermodynamic description of the Mg–Al–Mn system on the basis of critically assessed experimental data is presented. Particular attention is placed on the solubility of manganese in Mg-rich liquid alloys. The overall consistency between calculated and experimental phase equilibria is shown and the Calphad assessment enables a clear identification of consistency of various groups of data. In manganese-saturated liquid alloys the primary crystallizing phases are β-Mn and Al_8Mn_5 up to 23 wt.% Al and below 815 °C. Two invariant reactions, $\varepsilon(\text{hcp}) \leftrightarrow L + \beta\text{-Mn} + Al_8Mn_5$ (815 °C) and $L + \beta\text{-Mn} \leftrightarrow \text{Mg(hcp)} + Al_8Mn_5$ (637 °C) occur in Mg-rich liquid alloys. In addition, the solidus temperature and solid phase equilibria are given by the present thermodynamic calculation.

Keywords: Magnesium alloys; Manganese solubility; Calphad assessment; Phase diagrams

1. Introduction

The ternary Mg–Al–Mn system is one of the most important systems for the development of cast magnesium alloys. It is well known that an addition of a small amount of Mn results in improvement of corrosion resistance [1] and also grain refinement was observed early [2]. In order to clarify the mechanism for these effects and to exploit the potency of cast Mg alloys fully, a quantitative understanding is required regarding both equilibrium and non-equilibrium states of the alloys during solidification. To this end, a thermodynamic description consistent with the phase equilibria is needed. These data for Mg-rich alloys in the Mg–Al–Mn system provide crucial information for the development of this series of alloys and all alloys Mg–Al–Mn–*X* with *X* = Zn, Cu, Ni, Si like the AZ and AM series. Especially, the solubility of Mn in liquid has to be determined with the highest possible degree of accuracy and also the type of the precipitating phase is important to know.

There have been several experimental works on the solubility of manganese in liquid Mg–Al alloys [3–10]. However, some of them are inconsistent and the primary precipitates have not been clarified in most of the works. Furthermore, the type of reaction for the primary precipitates and the existence of invariant reactions in the Mg-rich corner are not clarified. Hence, in spite of practical importance, the phase diagram of the Mg-rich corner in the Mg–Al–Mn system has not been well established.

The Calphad approach is regarded as a powerful method to calculate the phase diagram and thermodynamic quantities for multi-component systems with the high accuracy demanded in development and designing of commercial alloys [11]. The thermodynamic description of Mg–Al–Mn system has been attempted earlier by focusing only on the Al-rich part [12] and that description is valid in fact only in the Al-rich part. The main objective of the present study is to obtain the thermodynamic description of the Mg–Al–Mn system with a high precision in the Mg-rich corner. In particular, our focus is placed on the description of the solubility of Mn in liquid with a high degree of accuracy as well as the proper description of phase equilibria over entire composition and temperature ranges.

In the next section, the experimental data for the phase equilibria reported so far are summarized and the reliability of these data is discussed. The thermodynamic models are explained in the third section. The calculated results and the comparison with the experimental data are presented in the fourth section. The detailed comparison in the solubility of Mn and the liquidus temperature clearly indicates inconsistencies in groups of experimental data. Furthermore, we present the overall consistency regarding liquidus surface over the entire composition range and also the solid-phase equilibria.

2. Experimental data in the literature

The Mg-rich phase equilibria in the Mg–Al–Mn system have been investigated in several works [3–10]. Beerwald [3] studied alloys with up to 7 wt.% Al and 5 wt.% Mn and showed that the Mn-solubility in liquid drastically decreases with increasing Al content in a temperature range between 660 and 800 °C. This behavior of Mn-solubility has been qualitatively confirmed by the other experimental works [5–7, 9, 10]. However, some of these experimental data are not quantitatively consistent with each other. As indicated by Thorvaldsen et al. [10], the measurement error becomes significantly large as the Al composition decreases and/or the heat treatment temperature increases. Furthermore, it should be noted that there exists a difficulty in determining the solubility because of iron contamination. It has been observed that in Mg–Al–Mn system, iron dissolves into Al–Mn binary phases by replacing Mn atoms [13] and stabilizes the Al–Mn binary phases [8]. Among the experimental data reported so far, the Mn-solubilities determined in [3], [5], and [10] are consistent with each other. Hence, these experimental data were the primary basis for the determination of thermodynamic parameters in the present study.

The precipitating phases in liquid in the Mg-rich corner were investigated by using a centrifuging technique [7–9].

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In these experimental works, it was observed that β-Mn, Al_8Mn_5, and $Al_{11}Mn_4$ phases are primary precipitates in the Mg-rich corner and isothermal sections for alloy composition range from 0 to 10 wt.% Al and from 0 to 3.5 wt.% Mn between 660 and 760 °C were constructed. Furthermore, the investigation of Thorvaldsen et al. [10] demonstrated that the precipitating phase is the Al_8Mn_5 phase for the investigated higher Al-content of Mg–Al liquid alloys.

In [5], the solidus temperature for alloys in the composition ranges 0–7.5 wt.% Al and 0–2.2 wt.% Mn was determined, based on both microstructural observations of the samples annealed for 8 or 16 h followed by quenching and mechanical tests of samples heated at a rate of 5 K/min. In the latter method, the solidus temperature was found to be the temperature at which rupture of the sample took place. The experimental results showed that the solidus temperature decreases with increase of Al composition.

Mirgalovskaya et al. [6] obtained vertical phase diagram sections at fixed Al composition between 800 and 200 °C and isothermal sections at 400 and 200 °C by means of microstructural observations and hardness tests of quenched samples. Also, Ageev et al. [4] presented isothermal sections at 430, 350, and 250 °C based on microstructural observations and micro-hardness tests of quenched samples. In these experimental works, however, some phases were not clearly specified. The solid-state solubilities suggested in these works [4, 6] will be discussed later in detail.

3. Thermodynamic description of the Mg–Al–Mn ternary system

In the present study, we employed thermodynamic parameters for the Al–Mg binary system assessed by Liang et al. [14] and, for the Al–Mn binary system, the work of Liu et al. [15] was used as a starting point. Thermodynamic assessment of the Mg–Mn system has been recently finished by Groebner et al. [16]. In this work, experimental results for the monovariant reaction $L' \rightarrow L + \delta\text{-Mn}$ were established and considered in the optimization and, simultaneously, the description of the Mg-rich part was improved. Their assessed parameters were employed in the present study. The solid phases and the models used in the present study are compiled in Table 1. It is noted that each of the phases fcc, hcp, and bcc occur in separate regions of the ternary system, thus ordinary phase names are also given for easier identification. The Gibbs energy functions for each phase are summarized in the following.

The Gibbs energy function, $G_i^{0,\varphi}(T) = G_i^{\varphi}(T) - H_i^{\mathrm{SER}}$, for the element i, in φ phase is given by,

$$G_i^{0,\varphi}(T) = a + b \cdot T + c \cdot T \ln T + d \cdot T^2 + e \cdot T^3 + f \cdot T^{-1} + g \cdot T^7 + h \cdot T^{-9} \quad (1)$$

where H_i^{SER} is the molar enthalpy of the stable element reference (SER) at 298.15 K and 1 bar, T is the absolute temperature. The Gibbs energy functions for Al, Mg, and Mn are taken from the SGTE compilation by Dinsdale [17].

The Gibbs energies for liquid, fcc, hcp, bcc, cbcc_A12 and cub_A13 are described by the substitutional solution model as follows,

$$G^{\varphi} = \sum_i x_i \cdot G_i^{0,\varphi} + RT \sum_i x_i \ln x_i + \sum_{i,j>i} x_i \cdot x_j \cdot \sum_v L_{i,j}^{v,\varphi} \cdot \left(x_i - x_j\right)^v \quad (2)$$

where $x_i(x_j)$ represents the mole fraction of element $i(j)$, R is the gas constant and $L_{i,j}^{v,\varphi}$ is the Redlich–Kister parameter representing the interaction between elements i and j [18]. The temperature dependence of $L_{i,j}^{v,\varphi}$ is usually described by linear function as $L_{i,j}^{v,\varphi} = A_{i,j}^{v,\varphi} + B_{i,j}^{v,\varphi} \cdot T$. In Eq. (2), the first and the second terms represent the Gibbs energies for mechanical mixture of elements and ideal mixing, respectively, and the third term corresponds to the excess Gibbs energy of mixing.

Note that in the original work for the Al–Mn system [15], the Gibbs energy of the bcc phase was described on the basis of the two-sublattice model in order to describe the disorder(bcc)/order(B2) transition. In the present study, however, we separately describe the Gibbs energies for bcc and B2 ordered phases because of the lack of experimental information about the solubility of Mg in B2 ordered phase.

The Gibbs energies for stoichiometric compounds such as $Al_{30}Mg_{23}$, Al_3Mg_2, $Al_{12}Mn$, Al_6Mn, Al_4Mn, $Al_{11}Mn_4$

Table 1. Solid phases of the Al–Mg–Mn system and the models used in the present study.

Phase model name	Ordinary phase name(s)	Pearson symbol/Prototype	Sublattices and constituents
Fcc	(Al), γ-Mn	*cF*4/Cu	(Al, Mg, Mn)
Hcp	(Mg), ε-AlMn	*hP*2/Mg	(Al, Mg, Mn)
Bcc	δMn, γ-AlMn	*cI*2/W	(Al, Mg, Mn)
Bcc_B2	γ′-AlMn	*cP*2/CsCl	$(Al, Mn)_1(Al, Mn)_1$
cbcc_A12	α-Mn	*cI*58/α-Mn	(Al, Mg, Mn)
Cub_A13	β-Mn	*cP*20/β-Mn	(Al, Mg, Mn)
AlMg-γ	γ-$Al_{12}Mg_{17}$	*cI*58/α-Mn	$(Mg)_{10}(Al, Mg)_{24}(Al, Mg)_{24}$
$Al_{30}Mg_{23}$	$Al_{30}Mg_{23}$	*hR*53/$Co_5Cr_2Mo_3$	$(Al)_{30}(Mg)_{23}$
Al_3Mg_2	Al_3Mg_2	*cF*1832/Al_3Mg_2	$(Al)_{140}(Mg)_{89}$
Al_8Mn_5	Al_8Mn_5	*hR*26/Al_8Cr_5	$(Al)_6(Mn)_2(Al, Mn)_5$
$Al_{12}Mn$	$Al_{12}Mn$	Unknown	$(Al)_{12}(Mn)$
Al_6Mn	Al_6Mn	*oC*28/Al_6Mn	$(Al)_6(Mn)$
Al_4Mn	Al_4Mn	Unknown (hexagonal)	$(Al)_4(Mn)$
$Al_{11}Mn_4$	$Al_{11}Mn_4$	*aP*15/$Al_{11}Mn_4$	$(Al)_{11}(Mn)_4$
AlMgMn-T	$Al_{18}Mg_3Mn_2$	*cF*184/$Al_{18}Cr_2Mg_3$	$(Al)_{18}(Mg)_3(Mn)_2$

phases are written as

$$G^{\mathrm{Al}_p M_q} = a^{\mathrm{Al}_p M_q} + b^{\mathrm{Al}_p M_q} \cdot T + \frac{p}{p+q} G^{0,\mathrm{fcc}}_{\mathrm{Al}} + \frac{q}{p+q} G^{0,\varphi'}_{M} \tag{3}$$

where M represents Mg or Mn and φ' corresponds to hcp or α-Mn depending on M = Mg or Mn, respectively. The ternary compound, $Al_{18}Mg_3Mn_2$, was also described as a stoichiometric compound and its parameters were taken from [12].

The Gibbs energies for the B2 ordered phase and γ-$Al_{12}Mg_{17}$ are described by the compound energy formalism (CEF) [19, 20] which can be generally expressed in the case of mixing on only 2 sublattices as

$$G^{\varphi} = \sum_{i,j} y^{\mathrm{I}}_i \cdot y^{\mathrm{II}}_j \cdot G^{0,\varphi}_{i:j} + \frac{p}{p+q} RT \sum_i y^{\mathrm{I}}_i \ln y^{\mathrm{I}}_i + \frac{q}{p+q} RT \sum_j y^{\mathrm{II}}_j \ln y^{\mathrm{II}}_j + {}^{\mathrm{E}}G^{\varphi} \tag{4}$$

where $y^{\mathrm{I}}_i(y^{\mathrm{II}}_j)$ is the site fraction of elemental constituents $i(j)$ on the sublattice I (II), p and q represent the fraction of lattice sites of sublattice I and II, respectively, $G^{0,\varphi}_{i:j}$ is the Gibbs energy of the formation of (mostly fictitious) compounds, $i_p j_q$. ${}^{\mathrm{E}}G^{\varphi}$ is the excess Gibbs energy expressed as follows,

$$\begin{aligned} {}^{\mathrm{E}}G^{\varphi} = {} & y^{\mathrm{I}}_{\mathrm{Al}} y^{\mathrm{I}}_{\mathrm{Mn}} \cdot y^{\mathrm{II}}_{\mathrm{Al}} \cdot L^{\varphi}_{\mathrm{Al,Mn:Al}} + y^{\mathrm{I}}_{\mathrm{Al}} y^{\mathrm{I}}_{\mathrm{Mn}} \cdot y^{\mathrm{II}}_{\mathrm{Mn}} \cdot L^{\varphi}_{\mathrm{Al,Mn:Mn}} \\ & + y^{\mathrm{II}}_{\mathrm{Al}} y^{\mathrm{II}}_{\mathrm{Mn}} \cdot y^{\mathrm{I}}_{\mathrm{Al}} \cdot L^{\varphi}_{\mathrm{Al:Al,Mn}} + y^{\mathrm{II}}_{\mathrm{Al}} y^{\mathrm{II}}_{\mathrm{Mn}} \cdot y^{\mathrm{I}}_{\mathrm{Mn}} \cdot L^{\varphi}_{\mathrm{Mn:Al,Mn}} \end{aligned} \tag{5}$$

where $L^{\varphi}_{\mathrm{Al,Mn}:M'}(L^{\varphi}_{M':\mathrm{Al,Mn}})$ represents the interaction parameter between Al and Mn on sublattice I (II) when the sublattice II (I) is occupied by M' (Al or Mn).

The Gibbs energy for Al_8Mn_5 involves mixing only on one sublattice and is given as follows [15],

$$\begin{aligned} G^{\mathrm{Al_8Mn_5}} = {} & y_{\mathrm{Al}} \cdot G^{0,\mathrm{Al_8Mn_5}}_{\mathrm{Al:Mn:Al}} + y_{\mathrm{Mn}} \cdot G^{0,\mathrm{Al_8Mn_5}}_{\mathrm{Al:Mn:Mn}} \\ & + \frac{5}{13} RT (y_{\mathrm{Al}} \ln y_{\mathrm{Al}} + y_{\mathrm{Mn}} \ln y_{\mathrm{Mn}}) \\ & + y_{\mathrm{Al}} \cdot y_{\mathrm{Mn}} \sum_{v=0}^{2} L^{v,\mathrm{Al_8Mn_5}}_{\mathrm{Al,Mn}} \cdot (y_{\mathrm{Al}} - y_{\mathrm{Mn}})^v \end{aligned} \tag{6}$$

where y_{Al} and y_{Mn} are the site fractions of Al and Mn on the third sublattice, respectively, $L^{v,\mathrm{Al_8Mn_5}}_{\mathrm{Al,Mn}}$ are the interaction parameters between Al and Mn on the third sublattice. $G^{0,\mathrm{Al_8Mn_5}}_{\mathrm{Al:Mn:Al}}$ and $G^{0,\mathrm{Al_8Mn_5}}_{\mathrm{Al:Mn:Mn}}$ are the Gibbs energies for formation of fictitious end members of the Al_8Mn_5-solution phase, "$Al_{11}Mn_2$" and "Al_6Mn_7", respectively, and these parameters are the main concern of the present calculation as discussed in the following.

For the Al–Mn system, we firstly used thermodynamic parameters assessed by Jansson [21] to calculate the present ternary system. However, it was observed that the calculated solubility of Mn in liquid was significantly different from the experimental data (by a factor of 2) and also the liquid compositions in the monovariant reaction for L + β-Mn + Al_8Mn_5 were inconsistent with experimental data. The latest assessment of the Al–Mn system corresponds to the one by Liu et al. [15] and our tentative calculation using their parameters showed considerable improvement regarding the monovariant reaction line for L + β-Mn + Al_8Mn_5. However, the discrepancy between the calculated and the experimental solubilities of Mn still remained.

The discrepancy appeared at the phase boundary for L/L + Al_8Mn_5 and, hence, was entirely ascribable to the thermodynamic description of Al_8Mn_5 phase in the Al–Mn binary system and the well established, essentially binary Mg–Al liquid phase. It is noted that the use of the ternary Redlich–Kister parameter makes no sense because the Mn-solubility in the liquid of interest is very small. Therefore, the thermodynamic parameters in Eq. (6) have to be modified and, then, it is necessary to reassess the entire Al–Mn phase diagram. However, the Al–Mn phase diagram calculated by [15] is in satisfactory agreement with the experimental results and the reassessment of the entire Al–Mn phase diagram is beyond the concern of the present study. We focused on the fact that below 750 °C, there are no experimental data to substantially support the equilibrium composition of the two-phase field for Al_8Mn_5 and β-Mn phases in the Al–Mn binary system. In the present study, hence, we attempted to reassess thermodynamic parameters for Al_8Mn_5 phase based on the experimental data of ternary solubility of Mn in liquid and, then, we employed a smoothing scheme of the Gibbs energy function to avoid the reassessment of the entire Al–Mn system, as explained in the following.

Our focus is placed on the Gibbs energies of fictitious end members, $Al_{11}Mn_2$ and Al_6Mn_7, and the parameters, $G^{0,\mathrm{Al_8Mn_5}}_{\mathrm{Al:Mn:Al}}$ and $G^{0,\mathrm{Al_8Mn_5}}_{\mathrm{Al:Mn:Mn}}$ in Eq. (6) were modified based on the experimental data for the ternary solubility of Mn in the liquid. The temperature range of this modification was below 750 °C because the experimental data for solubility at higher temperature are not reliable as mentioned in Section 2. In the present study, the modified parameters for the Gibbs energy functions were used below 755 °C, while the Gibbs energy functions given by [15] were used above 923 °C. Note that the temperature 923 °C corresponds to the peritectic reaction temperature for L + $Al_{11}Mn_4$ → Al_4Mn in the Al–Mn binary system. Hence, the calculated phase diagram above 923 °C is completely identical to the one by Liu et al. [15]. For the Gibbs energy between 755 and 923 °C, we introduced the smoothing function given as,

$$G^{0,\mathrm{Al_8Mn_5}}_{\mathrm{Al}_p\mathrm{Mn}_q}(T) = a^{p,q} + b^{p,q} \cdot T + d^{p,q} \cdot T^2 + e^{p,q} \cdot T^3 + \frac{p}{p+q} G^{0,\mathrm{fcc}}_{\mathrm{Al}} + \frac{q}{p+q} G^{0,\alpha-\mathrm{Mn}}_{\mathrm{Mn}} \tag{7}$$

It was also tried to use the term, $c^{p,q} \cdot T \ln T$, in Eq. (7), but acceptable results could not be obtained regarding the description of phase equilibria between 755 and 923 °C. The function given by Eq. (7) was appropriate to produce satisfying results as will be shown in Fig. 1. The function in Eq. (7) is subject to the following conditions,

$$G^{\mathrm{smooth}} = G^{\mathrm{modified}}, \quad \frac{\mathrm{d}G^{\mathrm{smooth}}}{\mathrm{d}T} = \frac{\mathrm{d}G^{\mathrm{modified}}}{\mathrm{d}T} \text{ at } 755\,^{\circ}\mathrm{C} \tag{8}$$

$$G^{\mathrm{smooth}} = G^{\mathrm{original}}, \quad \frac{\mathrm{d}G^{\mathrm{smooth}}}{\mathrm{d}T} = \frac{\mathrm{d}G^{\mathrm{original}}}{\mathrm{d}T} \text{ at } 923\,^{\circ}\mathrm{C} \tag{9}$$

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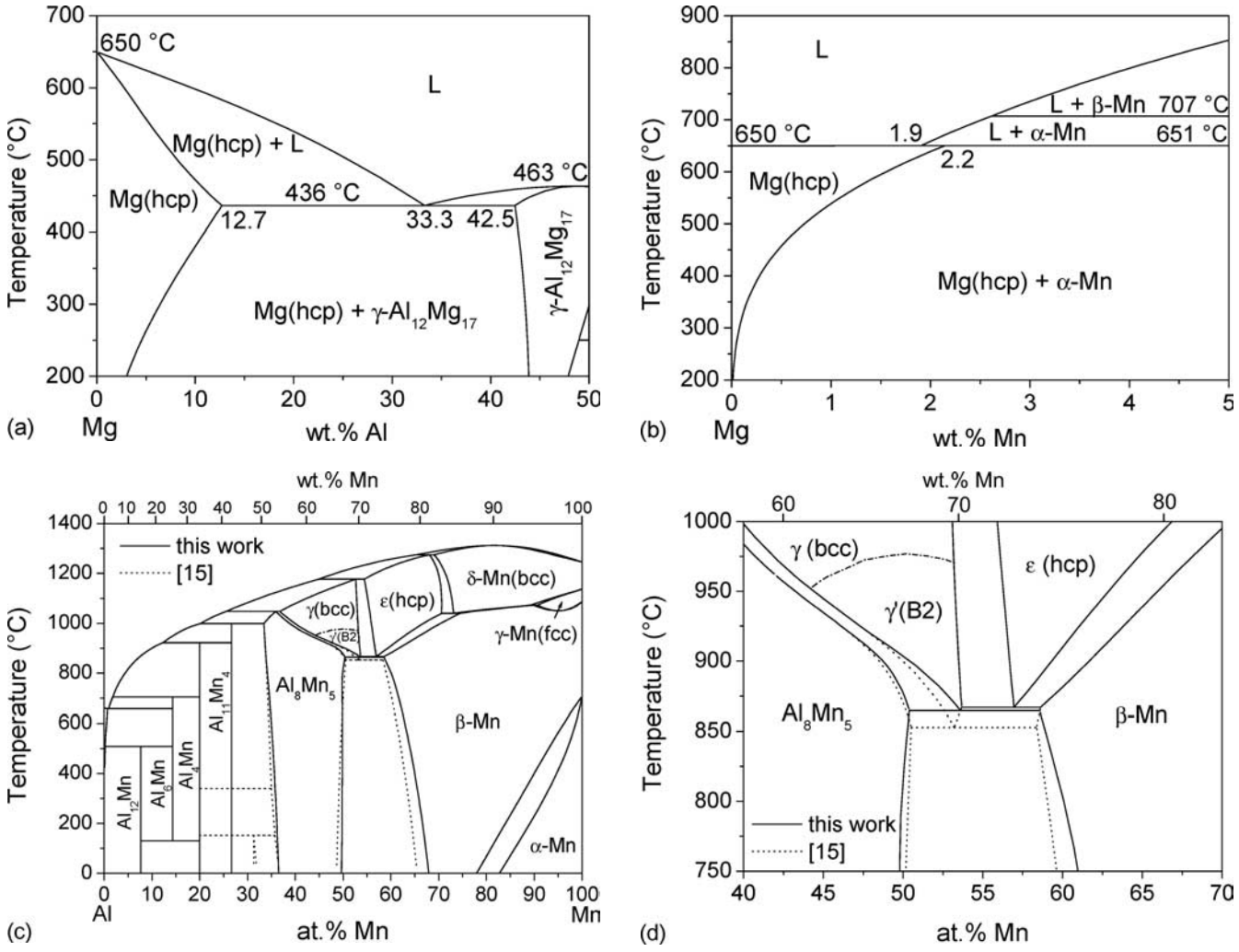


Fig. 1. (a) The Mg-rich part of phase diagram for the Al–Mg system [14]. (b) The Mg-rich part of phase diagram for the Mg–Mn system [16]. (c) The phase diagrams of Al–Mn system calculated by the present work (solid line) and in [15] (dashed line). (d) Enlargement of the phase diagram in the vicinity of eutectoid reaction, $\gamma' \rightarrow Al_8Mn_5 + \beta$-Mn. The dot-dashed curve indicates the disorder(bcc)-B2 transition temperature.

where G^{smooth} represents the smoothing function, $G^{modfied}$ corresponds to the Gibbs energy for fictitious end members assessed in the present study and $G^{original}$ is the original Gibbs energy function by [15]. The requirements given by Eqs. (8) and (9) assure that the Gibbs energies and entropies for fictitious phases vary smoothly with temperature at 755 and 923 °C. The parameters used in the present study are listed in Table 2. It is noted that $G^{modified}$ and $G^{original}$ display a linear temperature dependence, whereas G^{smooth} displays a cubic temperature dependence. In the present work, we chose the peritectic temperature, 923 °C, for the smoothing procedure. If we chose a lower temperature, e. g., 850 °C, the Gibbs energy varies sharply with temperature in the temperature range of the smoothing, which produces a bump of the phase boundary in the Al–Mn phase diagram. Our calculation showed that the peritectic temperature, 923 °C, is the most reasonable one in the preset smoothing scheme. However, our calculation also showed that phase equilibria in the Mg-rich corner of the Mg–Al–Mn system are not sensitive to the choice of the temperature range for the smoothing scheme.

In our calculation with the parameters given by Liu et al. [15] but using the Pandat software [22, 23], it can be shown that in the Al–Mn binary system, the $Al_{11}Mn_4$ phase decomposes into Al_8Mn_5 and Al_4Mn phases at low temperature and B2 ordered phase reappears at the lower temperature. These phase equilibria have not been experimentally observed and these are considered as a mathematical artifact. Hence, we performed a slight modification of the thermodynamic parameters for $Al_{11}Mn_4$ to remove these mathematical artifacts and then applied the same smoothing scheme as described above. So, for the AL_8Mn_5 and $Al_{11}Mn_4$ phases, we simply have three temperature ranges for thermodynamic parameters,

$G^{modified}$ $T < 755$ °C

G^{smooth} 755 °C $< T <$ 923 °C

$G^{original}$ $T >$ 923 °C

The parameters for $Al_{11}Mn_4$ employed in the present study are also given in Table 2. Our calculations showed that these smoothing schemes do not produce any discontinuity or visible bump of phase boundaries in the calculated binary or ternary phase diagrams. The strong curvature of the L + $Al_{11}Mn_4$ + Al_4Mn monovariant line in the Al-rich part will be discussed in the next section.

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Table 2. Thermodynamic parameters used in the present calculation. The parameters not listed here are the same as the ones given by Liu et al. [15].

Al_8Mn_5

298.15–1028.15 K (755 °C)
$G^{0,Al_8Mn_5}_{Al:Mn:Al} = 0.8461G^{0,fcc}_{Al} + 0.1539G^{0,\alpha-Mn}_{Mn} - 12\,662.3 + 4.6154 \cdot T$ $G^{0,Al_8Mn_5}_{Al:Mn:Mn} = 0.4615G^{0,fcc}_{Al} + 0.5385G^{0,\alpha-Mn}_{Mn} - 24\,796.5 + 5.3846 \cdot T$
1028.15–1196.15 K (923 °C)
$G^{0,Al_8Mn_5}_{Al:Mn:Al} = 0.8461G^{0,fcc}_{Al} + 0.1539G^{0,\alpha-Mn}_{Mn} + 82\,7551.6 - 2283.69 \cdot T + 2.066719 \cdot T^2 + 6.19 \cdot 10^{-4} \cdot T^3$ $G^{0,Al_8Mn_5}_{Al:Mn:Mn} = 0.4615G^{0,fcc}_{Al} + 0.5385G^{0,\alpha-Mn}_{Mn} - 34\,012.8 + 27.7691 \cdot T - 0.01739 \cdot T^2 + 4.22 \cdot 10^{-6} \cdot T^3$
1196.15–6000.00 K
$G^{0,Al_8Mn_5}_{Al:Mn:Al} = 0.8461G^{0,fcc}_{Al} + 0.1539G^{0,\alpha-Mn}_{Mn} - 12\,427.5 + 5.6923 \cdot T$ $G^{0,Al_8Mn_5}_{Al:Mn:Mn} = 0.4615G^{0,fcc}_{Al} + 0.5385G^{0,\alpha-Mn}_{Mn} - 23\,566.1 + 4.269 \cdot T$

$Al_{11}Mn_4$

298.15–1028.15 K
$G^{0,Al_{11}Mn_4} = 0.7333G^{0,fcc}_{Al} + 0.2667G^{0,\alpha-Mn}_{Mn} - 23800 + 6.0 \cdot T$
1028.15–1196.15 K
$G^{0,Al_{11}Mn_4} = 0.7333G^{0,fcc}_{Al} + 0.2667G^{0,\alpha-Mn}_{Mn} + 334\,141.9 - 945.026 \cdot T + 0.857784 \cdot T^2 + 2.57 \cdot 10^{-4} \cdot T^3$
1196.15–6000.00 K
$G^{0,Al_{11}Mn_4} = 0.7333G^{0,fcc}_{Al} + 0.2667G^{0,\alpha-Mn}_{Mn} - 22\,746.2 + 6.0 \cdot T$

4. Results and discussion

4.1. Binary systems

As mentioned in the previous section, in the present study, we employed thermodynamic parameters for the Al–Mg binary system assessed by Liang et al. [14] and, for the Mg–Mn binary system assessed by Groebner et. al. [16]. The Mg-rich parts of the phase diagram for these binary systems are given in Figs. 1a and b.

The calculated phase diagrams for the Al–Mn system after Liu et al. [15] and the present work are demonstrated in Fig. 1c. One can see exact agreement between both diagrams except for the following three features:

(i) As seen in Fig. 1c, in the phase diagram calculated with Pandat but using the parameters of Liu et al. [15], $Al_{11}Mn_4$ phase decomposes into Al_8Mn_5 and Al_4Mn phases at low temperature (340 °C, eutectoid) and then the B2 ordered phase appears at even lower temperatures and at about 31 at.% Mn (152 °C, peritectoid). Both reactions have not been experimentally observed and were not shown by Liu et al. [15]. These two mathematical artifacts are removed in the phase diagram calculated with the improved parameters from Table 2 in the present work.

(ii) There is difference in the equilibrium composition of the two-phase field for Al_8Mn_5 and β-Mn phases. However, there are no experimental data for the equilibrium composition of this two-phase field and the validity of the calculated phase boundary remains to be confirmed by future experiments.

(iii) The eutectoid temperature, $\gamma'(B2) \rightarrow Al_8Mn_5 + \beta\text{-Mn}$, is calculated to be 864 °C in the present work. This temperature is somewhat higher than the one calculated by Liu et al., 840 °C, as shown in Fig. 1d. As already mentioned, the main concern of the present study is the phase equilibria in the Mg-rich corner of the Mg–Al–Mn ternary system and this difference of the eutectoid temperature has no significant influence on the phase equilibrium in the Mg-rich corner. Hence, this disagreement is regarded to be acceptable in the present study.

Figure 2 shows calculated results of the enthalpy of formation and the absolute entropy for intermetallic compounds at 0 °C. The calculated results for enthalpy of formation and absolute entropy show fairly reasonable values. Note that the values for $Al_{12}Mn$, Al_6Mn, and Al_4Mn in this study and in [15] are identical. The calculated enthalpy of formation for Al_8Mn_5 at 50 at.% Mn is shown for the sake of the comparison with the corresponding experimental data [24]. The enthalpies of formation for Al_8Mn_5 and $Al_{11}Mn_4$ calculated by this work are slightly lower than the ones given by Liu et al. [15]. Note that this low enthalpy value for $Al_{11}Mn_4$ had to be introduced in order to remove the mathematical artifacts of the eutectoid $Al_{11}Mn_4$ decomposition and of the reappearance of ordered B2 below 152 °C as shown in Fig. 1c.

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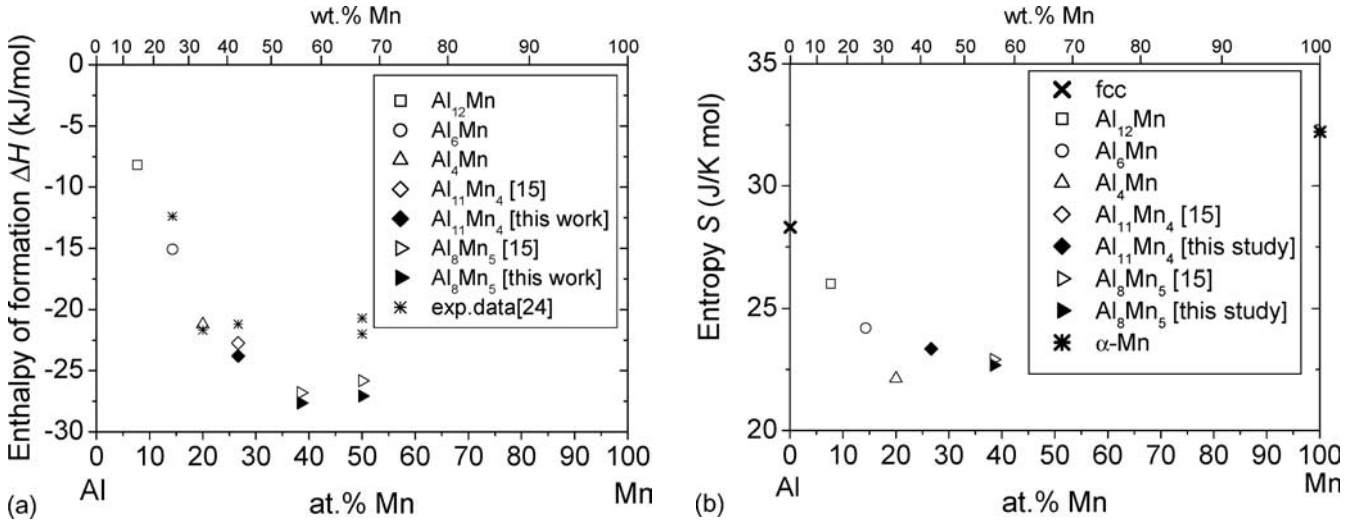


Fig. 2. (a) Enthalpy of formation and (b) absolute entropy for intermetallic compounds at 0 °C. The reference states in Fig. 2a are fcc-Al and α-Mn. Values for $Al_{12}Mn$, Al_6Mn, Al_4Mn, fcc and α-Mn in this study are identical to [15].

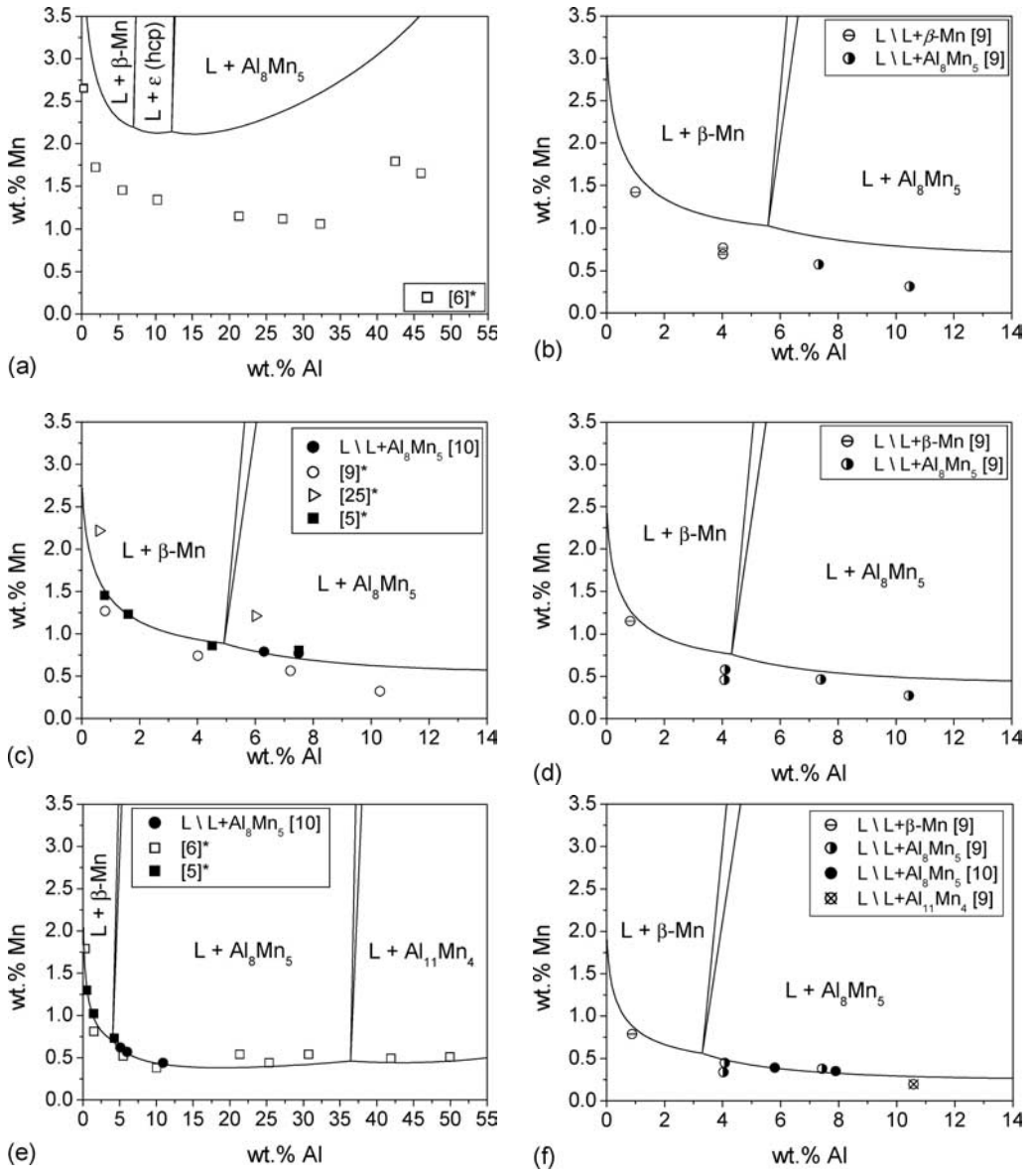


Fig. 3. Isothermal sections of Mg-rich corner calculated at various temperatures. (a) 850 °C, (b) 750 °C, (c) 730 °C, (d) 710 °C, (e) 700 °C, (f) 670 °C. All the experimental data represent the liquidus isotherm. The type of phase boundary is not specified in the experimental data marked by*. Note that the scales of horizontal axis in Figs. (a) and (e) are different from the other ones. The experimental data of [25] are extracted from a figure in [9].

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4.2. Solubilities and precipitates in Mg-rich ternary liquid alloys

In the following discussion, the composition of all ternary data is denoted in units of weight percent and all phase diagrams are obtained by thermodynamic calculation. The calculated isothermal sections at various temperatures are compared with experimental data in Fig. 3. The calculated solubilities are in good agreement with the experimental data at and below 730 °C, where the experimental accuracy is considered to be high. The calculated equilibrium solubility at 750 °C is somewhat higher than the experimental data of Simensen et al. [9]. It should be pointed out that in [9], the samples of the melt were taken by using crucibles after each heat treatment and such a sampling procedure might cause lower solubility values because the Mn composition in the liquid decreases due to precipitation of Mn-rich primary solid phase depending on the temperature drop during the sampling operation. Furthermore, one can see the considerable difference between the calculated result and experimental data of [6] at 850 °C. It is impossible to obtain agreement with these experimental data using any reasonable thermodynamic description. As mentioned in Section 2, the experimental data at high temperature is not reliable, which can be realized by the fact that the experimental solubility of [6] is much too small at the Mg–Mn binary edge, compared to the assessed and experimentally supported value of 4.9 wt.% Mn at 850 °C [16]. Note that these solubility data at 850 °C [6] were not considered in the current thermodynamic assessment.

In Fig. 3, the primary precipitates are specified in the experimental results of Simensen et al. [9] and Thorvaldsen et al. [10] and one can see that the calculated phase diagrams agree well with these experimental data.

The calculated solubility of Mn in liquid at two fixed Al compositions is shown in Fig. 4. These diagrams correspond to a portion of the vertical phase diagram sections at constant Al composition. The solid lines in Figs. 4a and b represent the phase boundaries between L/L + β-Mn and L/L + Al_8Mn_5, respectively. Note that in Fig. 4b, the L + Al_8Mn_5 phase field is replaced by the L + β-Mn phase field at high temperatures. The calculated solubility and primary phase are quite consistent with the experimental data.

There are many other experimental data for the solubility of Mn. The alloy compositions for all available data are shown in Fig. 5 with the calculated polythermal projection of liquidus surface. The thick lines indicate monovariant reaction lines and the thin lines represent the isotherms. The detailed comparison for each alloy sample between the calculated and experimental results is given in Fig. 6.

The liquidus temperature and the solubility of Mn in liquid are shown in Figs. 6a and b, respectively. The liquidus temperature was calculated at a given alloy composition. The solubility of Mn in liquid is calculated at a given temperature and Al composition. In both figures, the vertical axis represents the calculated value and the horizontal axis is the experimental value. The solid line is a visual aid, indicating perfect agreement of calculated and experimental data. One can readily realize the good agreement between the calculated results and the experimental ones [3, 5, 10]. From these comparisons, it can be clearly seen that an inconsistency exists in the experimental data of [6] at 850 °C, while their data at 700 °C are consistent with our

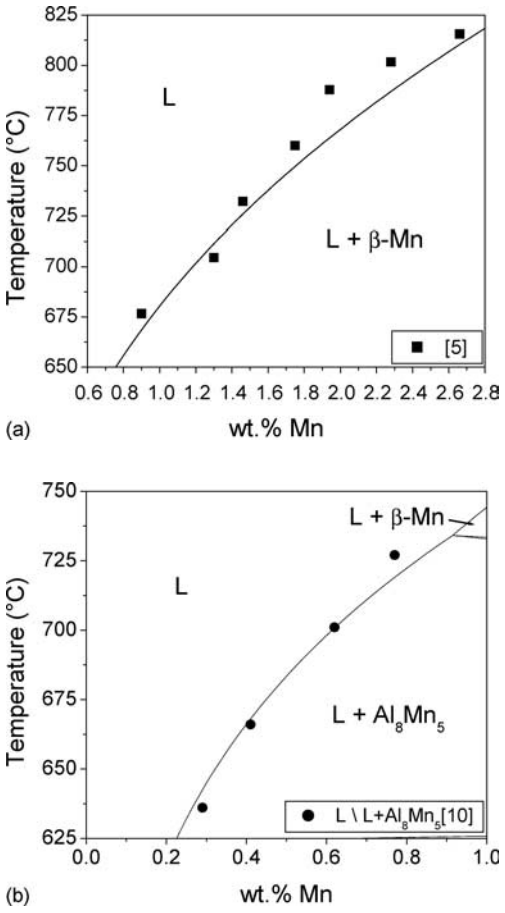


Fig. 4. Solubility of Mn in liquid at a fixed Al composition, (a) 0.8 wt.% Al, (b) 5.05 wt.% Al. Both the experimental data [5] and [10] represent the solubility of Mn in the liquid. The type of phase boundary is not specified in [5].

calculation as well as the other experimental data. Furthermore, the experimental data of Ageev et al. [4] always deviate largely from our calculated results and the other experimental data. This inconsistency is further supported by the fact that many of the samples of Ageev et al. [4] are even located on the primary precipitation field of Mg(hcp) in Fig. 5. This type of Calphad comparison allows us to identify groups of inconsistent experimental data clearly. This is a strong advantage over a simply graphically constructed set of phase diagrams, since each and every isolated data point is included in the joint comparison of Fig. 6. Furthermore, it enables us to see trends in individual groups of experimental data more clearly.

It should be pointed out that in Figs. 6a and b, as temperature increases, the calculated liquidus temperature becomes slightly lower than the experimental values and, correspondingly, as the solubility of Mn increases, the calculated value of the solubility becomes a little higher than the experimental one, even for the accepted groups of data. This deviation at the high temperature and high-Mn end of data is associated with the phase boundary between L/L + β-Mn (see Fig. 5). A better agreement regarding this phase boundary might be achieved by parameters stabilizing the β-Mn phase. However, the stabilization of β-Mn phase leads to a disagreement by shifting the liquid composition for L + β-Mn + Al_8Mn_5 monovariant reaction line and it is impossible to obtain complete thermodynamic con-

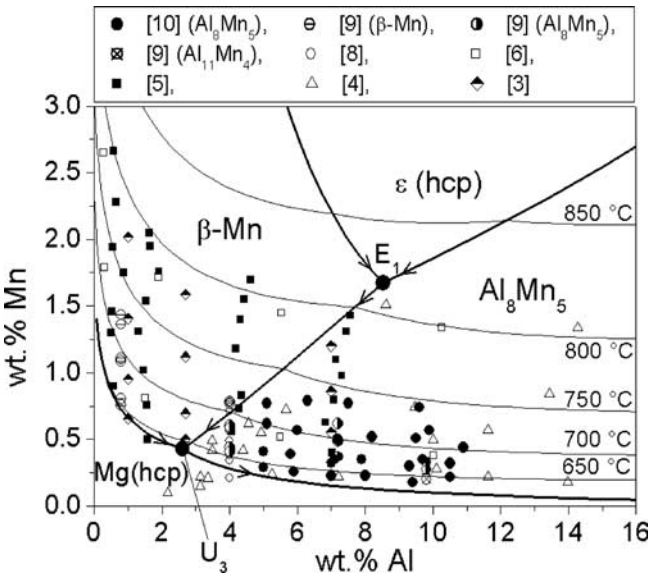


Fig. 5. Calculated liquidus surface, the thick lines indicate monovariant reaction lines and the thin lines represents the isotherms. Superimposed are the compositions of experimental alloys further compared to calculated data of the liquidus surface in Fig. 6. The primary solid phase as specified in experimental data of [10] and [9] is indicated in the legend. This information is not reported in the other work [3, 4, 5, 6, 8].

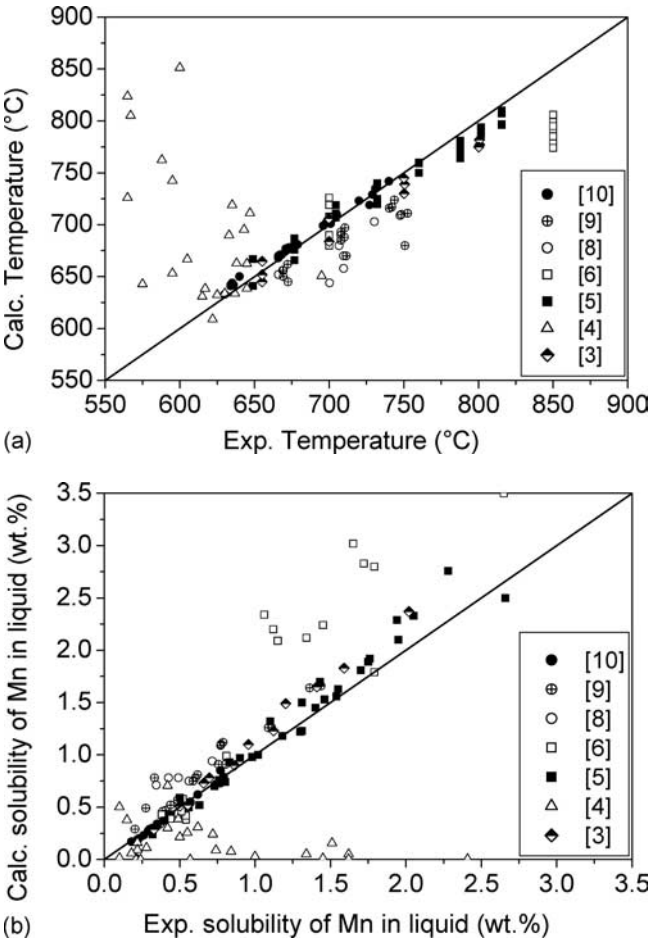


Fig. 6. Comparison between calculated results and experimental data for all alloy samples shown in Fig. 5. (a) Liquidus temperature at a given composition, (b) Solubility of Mn in liquid at a given temperature and Al composition. The straight line in Figs. (a) and (b) is a visual aid corresponding to perfect agreement between experimental values and the calculated results from the present thermodynamic model.

sistency for both the solubility values and the monovariant reaction line. In the present study, the thermodynamic parameter for the β-Mn phase was not changed, since the deviations regarding liquidus temperature and solubility shown in Fig. 6 are considered small and also because they occur at the high temperature end with the known larger experimental error.

From a closer look at the data of Simensen et al. [9], it is realized that their liquidus temperatures are slightly higher than the calculated values and their solubility of Mn has a slightly lower value than the calculated one. However, these slight differences may be acceptable. The more important information on the types of precipitating phases was carefully determined in [9], and this is quite consistent with our calculated result as will be shown in the following.

In Fig. 5, the experimental information about the primary precipitates are demonstrated [9, 10]. The more detailed comparison for the monovariant reaction, $L + \beta\text{-Mn} \leftrightarrow Al_8Mn_5$ is given in Fig. 7. It shows the projection of the liquidus surface onto the Mg–Al edge. The solid line corresponds to the calculated monovariant line for the reaction, $L + \beta\text{-Mn} \leftrightarrow Al_8Mn_5$. The same line is also seen in the projection of Fig. 5 between 815 °C and 637 °C. The experimental data represent equilibrium phases in liquid saturated with manganese [9]. In their work, the alloys were saturated with Mn by adding $MnCl_2$ at 780 °C followed by heat treatment for 1 or 2 h at various temperatures and, then, the particles in the quenched sample were analyzed by means of microprobe or X-ray diffraction (centrifuged sample). The calculated monovariant line is consistent with virtually all the experimental data. Note that in [8], both β-Mn and Al_8Mn_5 phases are detected in the melt corresponding to the bracketed point (4 wt.% Al, 710 °C). The $Al_{11}Mn_4$ precipitate observed in the 9.8 wt.% Al sample at 670 °C [9] cannot be reconciled with the equilibrium phase diagram and is also inconsistent with the data of [10] given in Fig. 5 at the same or even higher Al-content. The primary crystallization field of $Al_{11}Mn_4$ is calculated outside the frame of Fig. 5, beyond 23 wt.% Al.

Also, the composition of the precipitates was measured by using the microprobe analysis in the experimental works [8, 9]. These tie lines between liquid and the precipitate at 710 °C are given in Fig. 8 with the calculated isothermal section at that temperature. The dotted lines represent the calculated tie lines and the meshed areas correspond to three-phase regions. The qualitative agreement between the calculated and experimental tie lines is seen in this figure. However, it is noted that the measured compositions of the precipitates systematically have higher Al composition compared to the present calculation in Al–Mn binary edge. Especially, the measured composition of the β-Mn phase lies in the $\beta\text{-Mn} + Al_8Mn_5$ phase field. Furthermore, the comparison made at 730, 741, and 748 °C showed that the composition in the precipitates measured by Simensen et al. [9] also have a tendency of high Al composition. The calculated isothermal sections do not change significantly between 710 and 748 °C and are not shown here. It might be suggested that the accuracy of microprobe analysis of small precipitates could be inferior to those of binary Al–Mn data. These binary Al–Mn data are closely fixed by the thermodynamic data of β-Mn and Al_8Mn_5 phases, which are in turn linked to the phase boundaries in the Mg-rich alloys.

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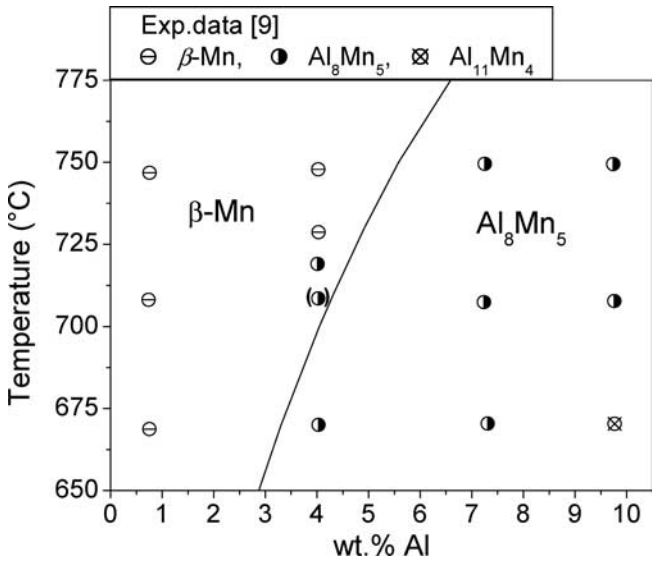


Fig. 7. Projection of the liquidus surface onto Mg–Al edge. The solid line indicates calculated monovariant line, L + β-Mn → Al_8Mn_5. The information on the Mn-content of this line, lost by the projection, is given in Fig. 5. The experimental data indicate which phase is in equilibrium with Mn-saturated liquid at the given temperature and Al-content. The bracketed point (4 wt.% Al, 710 °C) corresponds to both β-Mn and Al_8Mn_5 detected in melt [8]. The calculated range of Al_8Mn_5-precipitation is also supported by all the samples investigated by [10] as shown in Fig. 5.

In addition, the iron contamination of these samples was large. It has been observed that in the Mg–Al–Mn system, iron dissolves into Al–Mn binary phases by replacing Mn atoms [13], which leads to high Al ratio in the precipitate. Hence, it is considered that in [8] and [9], high Fe contamination of their sample might have caused the enhanced Al composition in the precipitates.

4.3. Liquidus surface of the Mg–Al–Mn system

Figure 9 demonstrates the polythermal projection of liquidus surface over the entire composition range. The thick lines represent monovariant lines and the dashed lines are the isotherms with an interval of 100 K. A blow-up of the Mg-rich corner was shown in Fig. 5. Four-phase reactions in the Mg-rich corner are labeled in Fig. 9. The present calculation shows the existence of the two following invariant reactions in the Mg-rich corner which are most important for Mg-rich alloys:

$\varepsilon(hcp) \leftrightarrow L + \beta\text{-Mn} + Al_8Mn_5$ (E_1 at 815 °C)

$L + \beta\text{-Mn} \leftrightarrow Mg(hcp) + Al_8Mn_5$ (U_3 at 637 °C)

The calculated compositions for each phase at the invariant reactions are given in Table 3. It is stressed that these data are very well established by the consistent thermodynamic description of all the experimental data.

The invariant reaction E_1, $\varepsilon(hcp) \leftrightarrow L + \beta\text{-Mn} + Al_8Mn_5$, at 815 °C requires a remelting of the ε(hcp)-phase. This is a rare form of the E-type reaction and due to the fact that the composition of ε(hcp) is located in the triangle of the formed phases. As seen in Table 3, ε(hcp) contains 1.2 wt.% Mg, much more than β-Mn + Al_8Mn_5. This is related to a particularity of the liquidus surface in Fig. 9: A large primary region of the Mn–Al-rich ε(hcp)-phase extends toward the Mg-rich corner, ending at this $\varepsilon(hcp) \leftrightarrow L + \beta\text{-Mn} + Al_8Mn_5$ invariant reaction with a liquid composition of 89.79 wt.% Mg. This ε(hcp)-phase is structurally identical to the Mg(hcp)-phase and is in fact described by the same thermodynamic model. The hcp phase demixes in the solid state into a Mg-rich and a Mg-poor phase and the latter, denoted as ε(hcp), contains a maximum of only 1.2 wt.% Mg in the stable phase equilibrium at 815 °C. Below 815 °C, this Mg-poor ε(hcp)-phase composition ceases to exist and is suppressed by the more stable Al–Mn phases, β-Mn, and Al_8Mn_5.

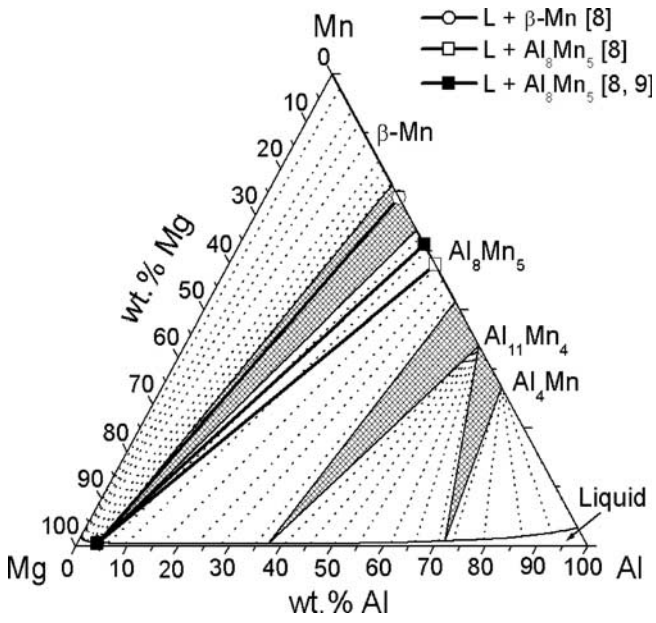


Fig. 8. Calculated isothermal section at 710 °C. The dotted lines represent tie lines and the meshed areas indicate three-phase regions.

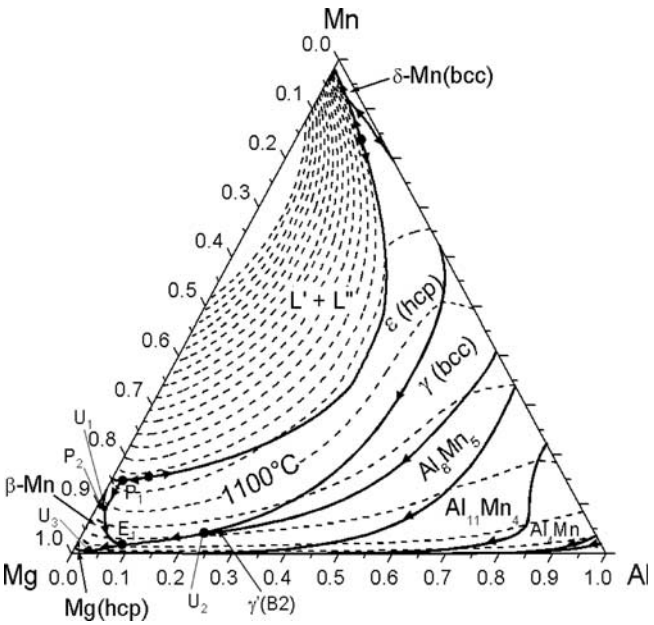


Fig. 9. Calculated liquidus surface. The solid lines represent monovariant lines, and the dashed lines are isotherms with an interval of 100 K. A blow-up of the Mg-rich corner is shown in Fig. 5.

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Table 3. Calculated invariant reactions at Mg-rich corner.

Reaction type	Temp (°C)	Phases	Composition (wt.%)		
			Al	Mg	Mn
$\varepsilon(hcp) \leftrightarrow L + \beta\text{-Mn} + Al_8Mn_5$	815	$\varepsilon(hcp)$	26.59	1.20	72.21
		L	8.54	89.79	1.67
		β-Mn	24.95	0.07	74.98
		Al_8Mn_5	32.98	0.00	67.02
$L + \beta\text{-Mn} \leftrightarrow Mg(hcp) + Al_8Mn_5$	637	L	2.61	96.96	0.43
		β-Mn	22.77	0.02	77.21
		Mg(hcp)	0.79	98.79	0.42
		Al_8Mn_5	33.08	0.00	66.92

The following phase equilibria are associated with this reaction E_1. Above this invariant temperature of 815 °C, there are three three-phase fields, $L + \varepsilon(hcp) + \beta\text{-Mn}$, $L + \varepsilon(hcp) + Al_8Mn_5$ and $\varepsilon(hcp) + \beta\text{-Mn} + Al_8Mn_5$. The three-phase field, $L + \varepsilon(hcp) + \beta\text{-Mn}$ results from an invariant reaction, $L + \delta\text{-Mn} \leftrightarrow \varepsilon(hcp) + \beta\text{-Mn}$, at 1064 °C indicated as U_1 in Fig. 9, while the three-phase field, $L + \varepsilon(hcp) + Al_8Mn_5$, originates from an invariant reaction, $L + \gamma'(B2) \leftrightarrow \varepsilon(hcp) + Al_8Mn_5$ at 902 °C indicated as U_2 in Fig. 9. The phase field, $\varepsilon(hcp) + \beta\text{-Mn} + Al_8Mn_5$, forms as a result of an almost degenerate invariant reaction, $\varepsilon(hcp) + \gamma'(B2) \leftrightarrow \beta\text{-Mn} + Al_8Mn_5$ which occurs in the vicinity of the Al–Mn binary edge just above 864 °C. The close competition of these four phases is already indicated in Fig. 1d by the almost direct contact of the two eutectoid binary reactions near 864 °C.

It is worthwhile to note another particularity regarding the liquidus surface shown in Fig. 9: The stable liquid miscibility gap of the Mg–Mn binary extends into the ternary up to 35 wt.% Al and down to a temperature which is calculated to be 1160 °C. It is noted that there are no experimental data for this concentrated ternary alloy region. The present calculation in Fig. 9 thus gives a very likely picture of the phase equilibria, but without direct experimental confirmation. The calculation shows a nonvariant three-phase equilibrium $\varepsilon(hcp) + L' + L''$. The temperature at this maximum is calculated to be 1265 °C and the compositions of L' and L'' are indicated by dots in Fig. 9 and the $\varepsilon(hcp)$-phase is Mn-rich (84.9 wt.% Mn, 12.5 wt.% Al, 2.6 wt.% Mg). Below this temperature, there exist two three-phase fields of $\varepsilon(hcp) + L' + L''$. One of them moves toward the Al-rich side with decreasing temperature and the other one moves toward the Al-poor side. The Al-rich one is closed at the critical point of $\varepsilon(hcp) + L$ at 1160 °C, while the Al-poor one disappears at an invariant reaction, $\varepsilon(hcp) + L' \leftrightarrow L'' + \delta\text{-Mn}$ at 1230 °C.

One can see in Fig. 9 that a strong curvature exists on the monovariant line for $L + Al_{11}Mn_4 \rightarrow Al_4Mn$ in the Al-rich corner. As mentioned in Section 2, the $Al_{11}Mn_4$ phase should have higher stability below 750 °C than assessed in [15], which leads to a wider primary phase field of $Al_{11}Mn_4$ in liquidus surface as seen in Fig. 9. This wide primary phase field in the ternary becomes the smaller phase field of $L + Al_{11}Mn_4$ at the binary Al–Mn edge and high temperature, above 923 °C. The strong curvature on the monovariant line cannot be avoided, in order to hold the thermodynamic consistency between the smaller phase field in the binary and the wide phase field in the ternary due to stabilization of $Al_{11}Mn_4$.

4.4. Phase equilibria of solid phases in Mg-rich alloys

As amply discussed above, the main focus of the present study is directed to the solubility of Mn in liquid Mg-rich alloys and it is seen from Figs. 3–9 that our thermodynamic description is very well consistent with the experimental data for these liquid phase boundaries. In the following, in order to demonstrate further consistency of the present thermodynamic description in the overall temperature range, the phase equilibria in the low temperature range are discussed.

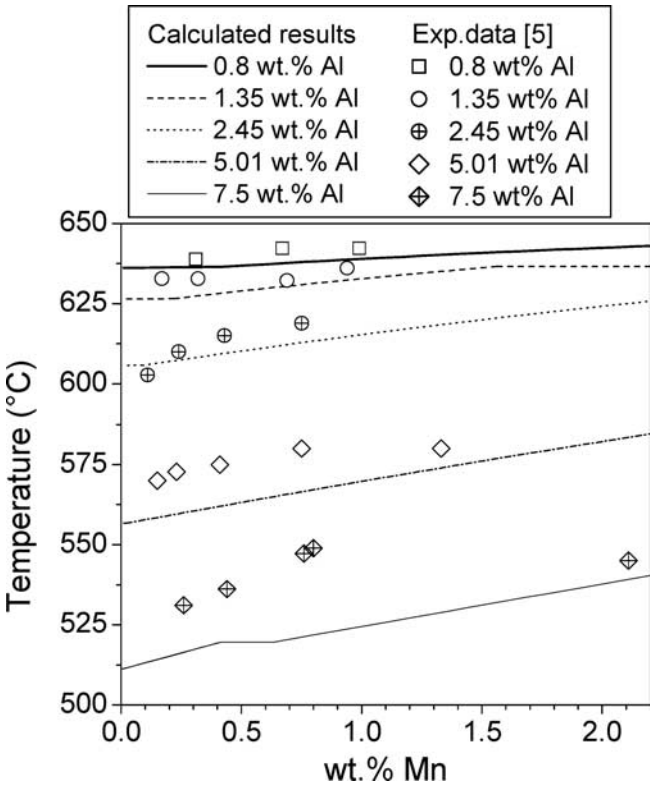


Fig. 10. Solidus temperature as a function of Mn composition at a fixed Al composition. The lines are from the present calculation. Below these lines the alloys are completely solid but may be multi-phase. The solid phases may vary, as seen by the bends in the curves.

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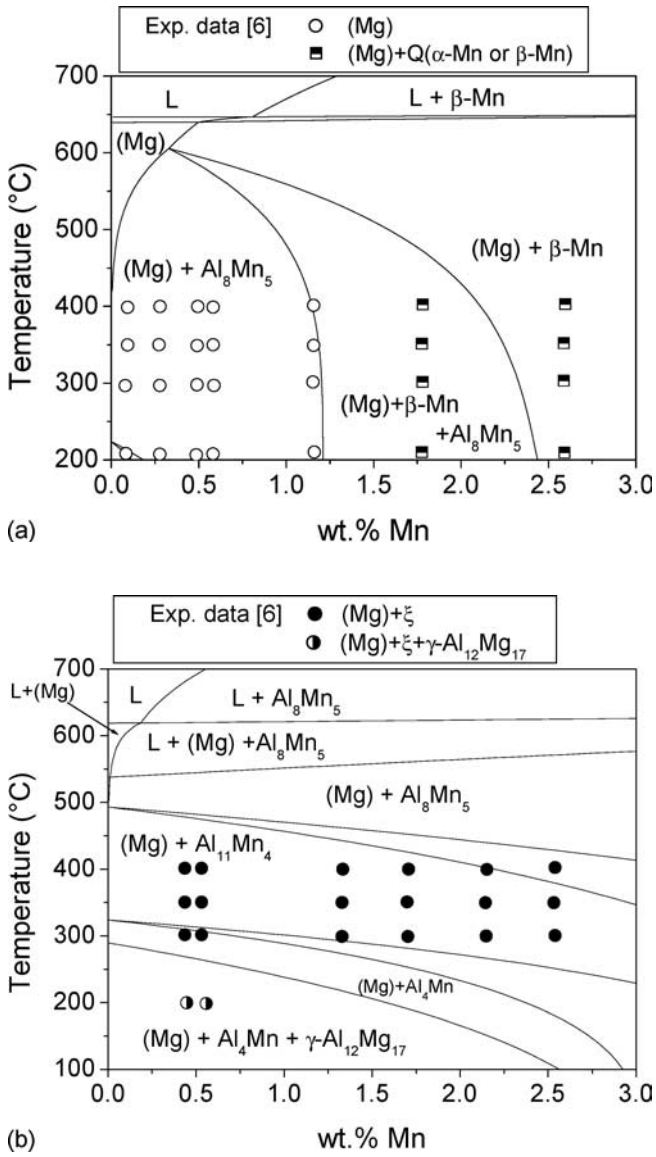


Fig. 11. Calculated vertical phase diagram sections at a fixed Al composition compared to experimental data, (a) 0.6 wt.% Al, (b) 6.0 wt.% Al. The Q- and ξ-phases are not clearly specified in [6]. For abbreviation, Mg(hcp) is labeled as (Mg).

4.4.1. Solidus temperature

Shown in Fig. 10 is the calculated solidus temperature as a function of Mn composition at various fixed Al contents with the corresponding experimental results [5]. Note that the alloys are completely solid below these lines, that is, the fraction of equilibrium liquid is zero along each line. However, the type of solid phases below these lines depends on the Mn composition and, also, the given Al composition. For instance, the calculated solidus line for 0.8 wt.% Al corresponds to the phase boundary between L + Mg(hcp)/Mg(hcp) phase fields at low Mn composition and it corresponds to the phase boundary between L + Mg(hcp) + β-Mn/Mg(hcp) + β-Mn phase fields at higher Mn composition (see also Fig. 11a with 0.6 wt.% Al). On the other hand, the calculated solidus for 5.01 wt.% Al represents the phase boundary between L + Mg(hcp) + Al_8Mn_5/Mg(hcp) + Al_8Mn_5 phase fields (see also Fig. 11b with 6 wt.% Al). The experimental data show that the solidus temperature increases with Mn composition and it decreases with increasing Al composition. The calculated results agree at least qualitatively well with the experimental ones. At higher Al composition, however, a quantitative disagreement is noted as discussed below.

As noted in Section 2, in the experimental work of [5], the solidus temperature was determined based on both microstructural observations and rupture tests. The experimental data shown in Fig. 10 represent the solidus temperature determined by the mechanical tests, however both the methods resulted in almost the same solidus temperature. It is obvious that there necessarily exists a critical amount of liquid to be detectable by these experimental methods and the critical amount of liquid is attained above the solidus temperature. Hence, the solidus temperature measured in [5] must be more or less higher than the equilibrium solidus temperature (zero phase fraction of liquid). The important point is that the difference between the equilibrium solidus temperature and the temperature for the detectable amount of liquid depends crucially on topology of phase boundaries just above the solidus temperature and, therefore, the accuracy of the measurement differs by the alloy composition. As an example, a temperature at which the mole fraction of liquid, f_{liquid}, equals to 0.08 was calculated based on the lever rule and plotted as a function of Mn content at a fixed Al composition in Fig. 12. One can see that the calculated temperatures for $f_{liquid} = 0.08$ agree even quantitatively well with all the solidus temperatures measured in [5]. The comparison of Figs. 10 and 12 reveals that the temperature for $f_{liquid} = 0.08$ approximately equals the

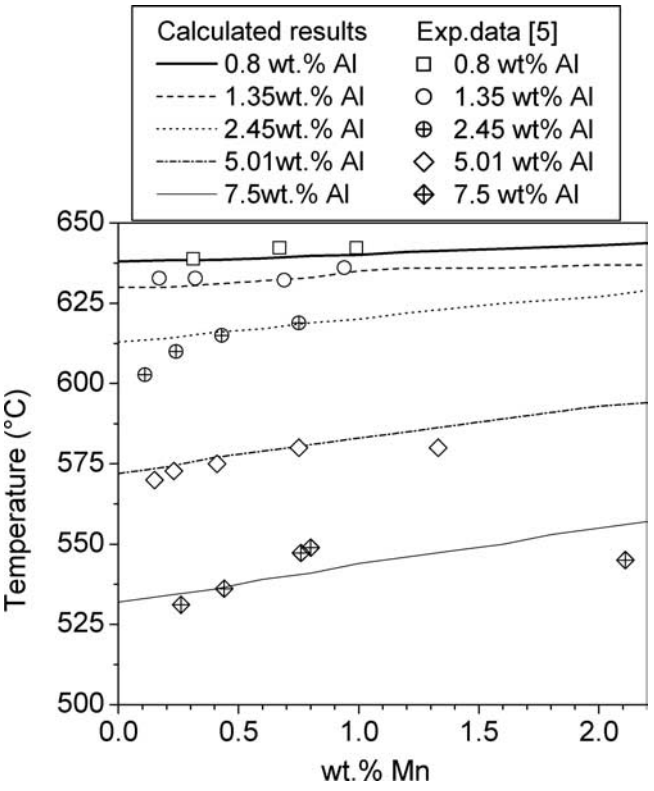


Fig. 12. Calculated temperature at which the fraction of liquid equals to constant value of 0.08, that might be assumed to be necessary for detection in the experimental method of Nelson [5]. The experimental data represent the solidus temperature and these are identical to the ones shown in Fig. 10.

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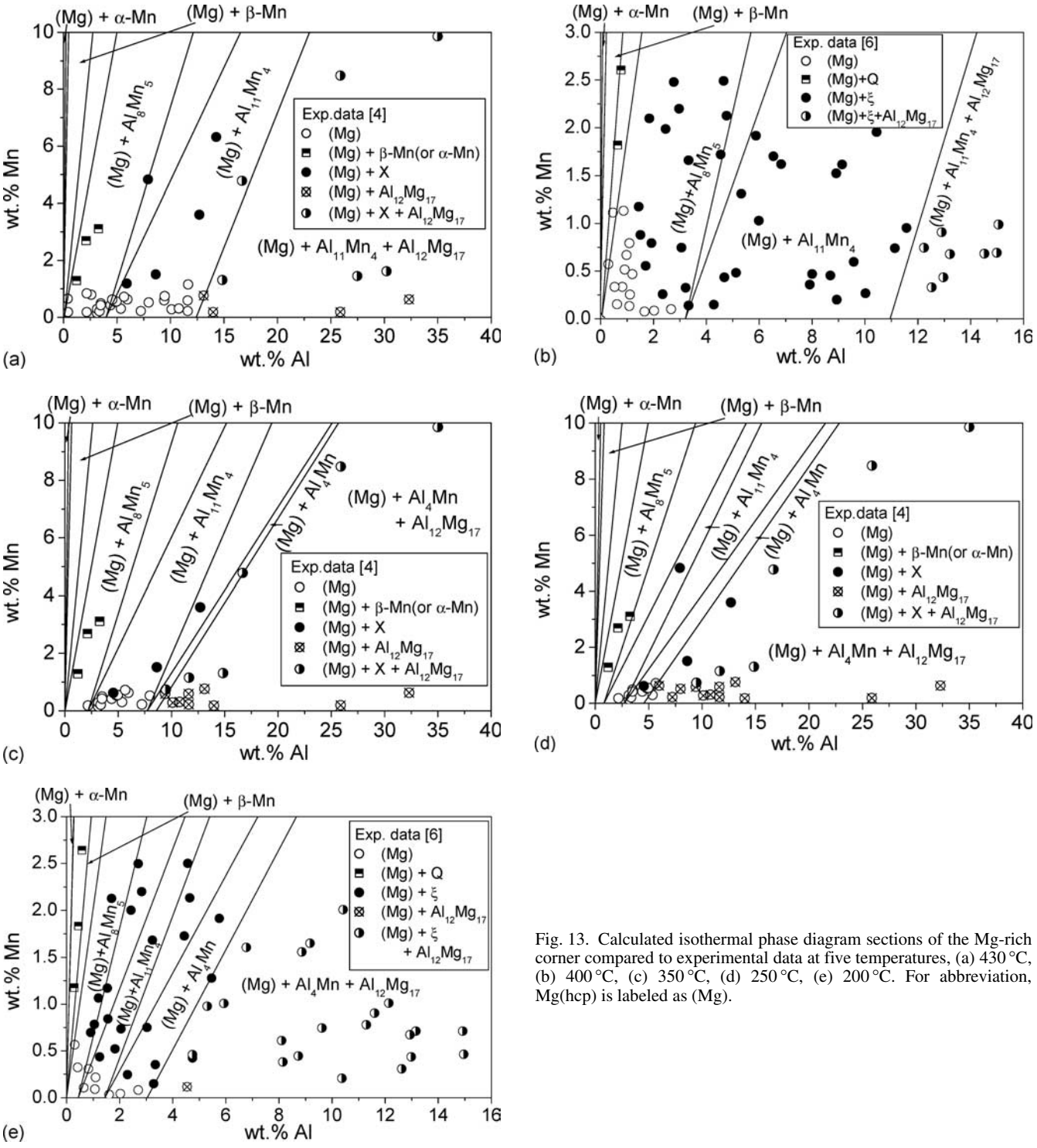


Fig. 13. Calculated isothermal phase diagram sections of the Mg-rich corner compared to experimental data at five temperatures, (a) 430 °C, (b) 400 °C, (c) 350 °C, (d) 250 °C, (e) 200 °C. For abbreviation, Mg(hcp) is labeled as (Mg).

calculated solidus temperature ($f_{liquid} = 0$) at low Al composition. However, at high Al composition, the temperature for $f_{liquid} = 0.08$ is considerably different from the equilibrium solidus temperature. This probably explains the difference appearing at high Al composition in Fig. 10.

4.4.2. Solid-state equilibria

The calculated vertical sections at a fixed Al composition are given in Fig. 11 with the experimental data representing the type of equilibrium phases observed at a given temperature and composition. In the experimental work of Mirgalovskaya et al. [6], the equilibrium phases were determined based on microstructural observations and microhardness tests. In their work, a Q-phase was referred to as a cubic Mn phase and a ξ-phase was considered as a phase appearing in the Al–Mn binary system. In Fig. 11a, the calculated phase fields involving β-Mn(cubic) phase agree well with the appearance of Q-phases. In Fig. 11b, the Mg(hcp) + $Al_{11}Mn_4$ phase field is quite consistent with the experimentally determined phase field of Mg(hcp) + ξ and also the calculated result is in agreement with the experimental data regarding the appearance of γ-$Al_{12}Mg_{17}$.

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One can see that in Fig. 11a, the solubility of Mn in Mg(hcp) claimed in [6] is much larger than the calculated one. A huge single-phase Mg(hcp) field, up to 1.2 wt.% Mn below 400 °C, is claimed in [6], which is even larger than the one in the binary Mg–Mn edge (e. g. 0.27 wt.% Mn at 400 °C). Contrary to that, in the ternary system with the appearance of the stable phase Al_8Mn_5, the solubility must be smaller. Therefore, the large solubility measured in [6] cannot be thermodynamically justified. The temperature independence of the claimed single-phase field is also against thermodynamic relations. The small amount of Al_8Mn_5 precipitates might have been below the detection limit.

The calculated isothermal sections at various temperatures are demonstrated in Fig. 13 with the experimental data indicating the type of equilibrium phase at a given temperature and composition. These equilibrium phases were determined based on the microstructural observations and microhardness tests [4, 6]. The X-phase was referred to as Mn-rich phase in the experimental work [4]. In the calculated isothermal section, the Mn-rich phases appear at high Mn composition, and the calculated isothermal sections are quite consistent with these experimental results, provided that Al–Mn binary phases such as Al_8Mn_5, $Al_{11}Mn_4$, Al_4Mn were not distinguishable by microstructural observations and hardness tests in [6] and [4].

In Fig. 13, the experimentally claimed Mg(hcp)-single-phase field is extremely large and, furthermore, the two-phase samples of Mg(hcp) + γ-$Al_{12}Mg_{17}$ have a high Mn composition without Mn-rich precipitate. These high Mn compositions cannot be thermodynamically justified as discussed in detail with Fig. 11a. Again, the order of magnitude of manganese claimed to be dissolved in Mg(hcp) is up to about 1 wt.% Mn in Fig. 13, whereas the consistent thermodynamic calculation indicates vanishing Mn solid solubility for all these alloys.

5. Conclusion

In the present study, a thermodynamic description for the Mg–Al–Mn system was accomplished by refining the binary Al–Mn data, based on the experimental data mainly for solubility of Mn in liquid Mg-rich alloys. Detailed comparisons between the calculated phase diagrams and experimental results were demonstrated. The present calculation based on the consistent thermodynamic description demonstrates the following important results;

(1) Mn solubility in liquid Mg alloys drastically decreases by small Al-additions.
(2) Beyond the solubility limit the primary crystallizing phases are β-Mn and Al_8Mn_5 up to 23 wt.% Al and below 815 °C. The limiting line between these two precipitates is the monovariant reaction L + β-Mn + Al_8Mn_5, that is well established by the thermodynamic calculation. A significant decrease in Al-content of that liquid occurs between 815 °C and 637 °C.
(3) The following two invariant reactions occur in Mg-rich liquid alloys,
ε(hcp) ↔ L + β-Mn + Al_8Mn_5 (at 815 °C)
L + β-Mn ↔ Mg(hcp) + Al_8Mn_5 (at 637 °C)
where ε(hcp) and Mg(hcp) have the same hcp structure, but distinctly different compositions.
(4) The Calphad technique enables a clear identification of groups of consistent or inconsistent experimental data, since each and every isolated data point is included in the joint comparison. Furthermore, it elucidates trends in individual groups of experimental data more clearly.
(5) The solidus and solid state phase equilibria are also given by the consistent thermodynamic description. This enables a quantitative thermodynamic treatment of solidification of Mg-rich ternary Mg–Al–Mn alloys using equilibrium or Scheil models.

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References

[1] J.D. Hanawalt, C.E. Nelson, J.A. Peloubet: Trans. AIME 147 (1942) 273.
[2] C.E. Nelson: Trans. Am. Foundrymen's Soc. 56 (1948) 1.
[3] A. Beerwald: Metallwirtschaft 23 (1944) 404.
[4] N.V. Ageev, I.I. Kornilov, A.N. Khlapova: Izv. Sekt. Fiz-Khim. Anal. 16 (1948) 130.
[5] B.J. Nelson: J. Metals 3 (1951) 797.
[6] M.S. Mirgalovskaya, L.N. Matkova, E.M. Komova: Trudy Inst. Met. Im. A.A. Baikova, Akad. Nauk 2 (1957) 139.
[7] B.C. Oberlander, C.J. Simensen, J. Svalestuen: Conf. Magnesium Technology, Inst. of Metals (1986) 133.
[8] C.J. Simensen. B.C. Oberlander, J. Svalestuen, A. Thorvaldsen: Z. Metallkd. 79 (1988) 537.
[9] C.J. Simensen, B.C. Oberlander, J. Svalestuen, A. Thorvaldsen: Z. Metallkd. 79 (1988) 696.
[10] A. Thorvaldsen, C.A. Aliravci: Adv. Prod. Fabr. Light Met. Met. Matrix Comp. Proc. Int. Symp. (1992) 277.
[11] Y.A. Chang, S.-L. Chen, F. Zhang, X.-Y.F.Y. Xie, R. Schmid–Fetzer, W.A. Oates: Progr. Mater. Science 49 (2004) 313.
[12] I. Ansara: Al–Mg–Mn system in COST 507, Thermochemical database for light metal alloys, Vol. 2 (1998) 325.
[13] N.F. Lashko, G.I. Morozova, I.Y. Mukhina, M.A. Timonova: Russ. Metall. 4 (1971) 138.
[14] P. Liang, H.-L. Su, P. Donnadieu, M.G. Harmelin, A. Quivy, P. Ochin, G. Effenberg, H.-J. Seifert, H.L. Lukas, F. Aldinger: Z. Metallkd. 89 (1998) 536.
[15] X.J. Liu, I. Ohnuma, R. Kainuma, K. Ishida: J. Phase Equilibria 20 (1999) 45.
[16] J. Groebner, D. Mirkovic, M. Ohno, R. Schmid–Fetzer: J. Phase Equilibria Diffusion, submitted.
[17] A.T. Dinsdale: Calphad 15 (1991) 317.
[18] O. Redlich, A.T. Kister: Ind. Eng. Chem. 40 (1948) 345.
[19] J.O. Andersson, A. Fernandez Guillermet, M. Hillert, B. Jansson, B. Sundman: Acta metall. 34 (1986) 437.
[20] M. Hillert: J. Alloys and Comp. 320 (2001) 161.
[21] A. Jansson: Metall. Trans. A 123 (1992) 2953.
[22] Pandat-Phase Diagram Calculation Engine for Multicomponent Systems, CompuTherm LLC, 437 S. Yellowstone Dr., Suite 217, Madison, Wisconsin, USA (2000).
[23] S.-L. Chen, S. Daniel, F. Zhang, Y.A. Chang, W.A. Oates, R. Schmid–Fetzer: J. Phase Equilibria 22 (2001) 373.
[24] O. Kubaschewski, G. Meymer: Trans. Faraday Soc. 6 (1960) 473.
[25] E.F. Emiley: Principles of Magnesium Technology, Pergamon Press, Oxford (1966).

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Correspondence address

Professor Dr. Rainer Schmid–Fetzer
Clausthal University of Technology, Institute of Metallurgy
Robert-Koch-Str. 42, D-38678 Clausthal-Zellerfeld, Germany
Tel.: +49 5323 72 2150
Fax: +49 5323 72 3120
E-mail: schmid-fetzer@tu-clausthal.de

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